1. Why lift sets repel linear structure: Kloosterman moments

Study `research/lift_repulsion_mechanism_20261027/`, 2026-10-27.

- Run: `reports/run-20261027T1459Z/` (748 sets, exact counts), master seed `0x11F7_2026_1027_C` (the parent's, so the `orig` population is the parent's).
- Protocol: `PROTOCOL.md`, frozen before the run; sha256 of protocol, source and `Cargo.toml` in `FREEZE.sha256` (verified after the run).
- Harness: `src/main.rs`, std-only Rust. Field arithmetic, lift set, null generator and the count-mode subspace DFS are copied from `research/lift_dense_factor_base_20261027/src/main.rs`; the parent's 104 `orig` ratios are reproduced with max abs difference 0.0.
- Plotting: `plot.py`, layout only.

Each statement is tagged **theorem**, **exact computation**, **measured** or **not established**.

1.1. Question and short answer

Parent finding (measured): on $E : y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^N}$, N prime, the set L of lifting abscissae contains fewer all-lift k-dimensional $\mathbb{F}_2$-subspaces than a uniform random set of equal size, and its count scatters less. At $k = 4$ the normalised counts are 0.68, 0.92, 0.98 at $N = 9, 11, 13$.

Hypothesis tested: the cause is the Kloosterman (Sato–Tate) fourth moment of the Walsh spectrum, 2 instead of the Gaussian 3, which gives fewer additive quadruples and fewer all-lift subspaces.

Verdict. The premise is **a theorem**: every curve has the same Walsh power sums ($\sum W^4 = 2q^3 - 2q^2 - 3q$, normalised 4th moment $2 - O(1/q)$), so curves have exactly the same, far-below-random, number of additive quadruples. A first-order expansion of the subspace counts in the 3-point and 4-point correlations is exact at $k = 2$ and explains about 70 % of the $k = 4$ curve deficit. The mean shift comes from the 4th moment, the $\mathrm{Tr}(a)$-dependent offset from the 3rd, and the smaller scatter from the 3rd moment (not the 4th). About 30 % of the odd-N deficit is **not established**. Composite $N = 12$ is not explained by subfield cancellation.

id	rule	verdict	numbers
I1, I2	exact Walsh 4th and 3rd power sums on every curve	**pass**	120/120 curves
P1	curves have lower Walsh 4th moment	**pass**	2.0010 vs 2.980 at N = 11; d = 13.7; p = 1e-5
P2	ratio_4 rises with the 4th moment, pooled and within nulls	**fail**	within nulls: 0.195 / 0.029 / −0.118 (N = 9 / 11 / 13)
P3a	$A_2 = E_2 + D_{\mathrm{line}}$ exactly	**pass**	exact for all 748 sets
P3b	first-order $k = 4$ within factor 2, $R^2 \geq 0.5$	**pass**	pred/obs 0.885 / 0.705 / 0.683; R^2 0.61 / 0.92 / 0.96
PC	planted Walsh bias raises the count	**pass**	Spearman 0.944 / 0.998 / 0.994

Table 1: Preregistered verdicts, computed by the program (`summary.json` → `verdicts`). P2 fails by the rule, as predicted in the protocol.

1.2. Derivation

Throughout, $q = 2^N$, $u = q - 1$, $c = \sqrt{b} \neq 0$, $\varepsilon = (-1)^{\mathrm{Tr}(a)}$. Put $\psi(x) = (-1)^{\mathrm{Tr}(x + a + b/x^2)}$ for $x \neq 0$ and $\psi(0) = 0$. Then $\psi = +1$ exactly on the lift set L and -1 on $\mathbb{F}_q^* \setminus L$. Write $W(w) = \sum_x \psi(x)(-1)^{w \cdot x}$ (the bit dot product is a non-degenerate form, so every power sum below equals the one for $\mathrm{Tr}(wx)$).

Tag: theorem (proved here by hand; each identity is also checked as exact integer equality on every curve of the run).

1.2.1. Walsh spectrum is a Kloosterman spectrum

Since $\mathrm{Tr}(b/x^2) = \mathrm{Tr}(c/x)$,

$$\psi(x) = \varepsilon(-1)^{\mathrm{Tr}(x + c/x)}, \quad W(w) = \varepsilon K(1 + w; c), \quad K(t; c) = \sum_{x \neq 0} (-1)^{\mathrm{Tr}(tx + c/x)}.$$

The substitution $x \to x/t$ gives $K(t; c) = K(1; tc)$, so as t runs over $\mathbb{F}_q^*$ the pair (t, c) runs over every Kloosterman sum $K(\lambda) := K(1; \lambda)$ exactly once. $K(0; c) = -1$. All curves with the same N therefore share one Walsh spectrum up to translation $w \to w + 1$ and the global sign ε.

1.2.2. Power sums of the Walsh spectrum

Proposition 1. For every N, a and $b \neq 0$:

1. $\sum_w W(w)^2 = qu$ (Parseval; true for every ψ with $\psi^2 = 1_{x \neq 0}$).

2. $\sum_w W(w)^3 = qC_3$ with $C_3 = -\varepsilon(q-2)$ for N odd and $C_3 = \varepsilon(q+2)$ for N even.
3. $\sum_w W(w)^4 = q(2q^2-2q-3)$.

Proof of 2. $\sum_w W^3 = q\sum_{x+y+z=0}\psi(x)\psi(y)\psi(z)$. For $x,y\neq 0, x\neq y$ and $z=x+y$ the linear traces cancel, $\mathrm{Tr}(x)+\mathrm{Tr}(y)+\mathrm{Tr}(x+y)=0$, so the product is $\varepsilon(-1)^{\mathrm{Tr}(c(1/x+1/y+1/(x+y)))}$. Put $y=tx$ with $t\in\mathbb{F}_q\setminus\{0,1\}$: the exponent is $\mathrm{Tr}(ch(t)/x)$ with $h(t)=(t^2+t+1)/(t(t+1))$. Summing over $x\neq 0$ gives -1 when $h(t)\neq 0$ and $q-1$ when $h(t)=0$. h vanishes exactly at the primitive cube roots of unity, which lie in $\mathbb{F}_q$ iff N is even. Odd N: $\sum_t = -(q-2)$. Even N: $2(q-1)-(q-4) = q+2$. $\square$

Proof of 3. $K(t;c)^2 = \sum_s (-1)^{\mathrm{Tr}(ts)}G(s)$ with $G(s)=\sum_{x+y=s,xy\neq 0}(-1)^{\mathrm{Tr}(c(1/x+1/y))}$. Then $G(0)=u$. For $s\neq 0$ put $x=sz$, $z\notin\{0,1\}$: $G(s)=\sum_z(-1)^{\mathrm{Tr}(\lambda/(z^2+z))}$ with $\lambda=c/s$. The map $z\mapsto z^2+z$ is two-to-one from $\mathbb{F}_q\setminus\{0,1\}$ onto $\{v\neq 0:\mathrm{Tr}(v)=0\}$, and $2[\mathrm{Tr}(v)=0]=1+(-1)^{\mathrm{Tr}(v)}$, so

$$G(s)=\sum_{v\neq 0}(-1)^{\mathrm{Tr}(\lambda/v)}+\sum_{v\neq 0}(-1)^{\mathrm{Tr}(v+\lambda/v)} = -1+K(\lambda).$$

Parseval on $t\in\mathbb{F}_q$ gives $\sum_t K(t;c)^4 = q\sum_s G(s)^2 = q\Big(u^2+\sum_{\lambda\neq 0}(K(\lambda)-1)^2\Big)$. Now $\sum_{\lambda\neq 0}K(\lambda) = \sum_{v\neq 0}(-1)^{\mathrm{Tr}(v)}\sum_{\lambda\neq 0}(-1)^{\mathrm{Tr}(\lambda/v)} = 1$ and $\sum_{\lambda\neq 0}K(\lambda)^2 = qu-1$ (part 1 with the $t=0$ term removed). Hence $\sum_{\lambda\neq 0}(K-1)^2 = qu-1-2+u = q^2-4$ and $\sum_w W^4 = q(u^2+q^2-4) = q(2q^2-2q-3)$. $\square$

Consequences. For a curve the normalised 4th moment is $\kappa = \sum W^4/(qu^2) = 2-O(1/q)$, identical on every curve: the Sato–Tate value 2, with no scatter. A random ± 1 function has $\kappa = 3$ (the Gaussian value; exactly $3-O(1/q)$ for a uniform balanced one). The same power sums, written as correlations of ψ over linear relations ($x+y+z+w=0$ etc.), are:

$$C_3 = \sum_{x+y+z=0}\psi\psi\psi = -\varepsilon(q-2), \quad R_4 = \sum_{x+y+z+w=0}\psi\psi\psi\psi = 2q^2-2q-3.$$

The R_4 sum contains degenerate solutions ($x=y$ and $z=w$, and the two other pairings): $3u^2-2u$ of them, each contributing $+1$. The sum over four distinct nonzero elements is therefore

$$C_{4'} = R_4 - 3u^2 + 2u = -(q-2)(q-4).$$

A random sign pattern has $EC_{4'}=0$ and sd $\sim q^{3/2}$. So a curve has $C_{4'}\sim -q^2$, which is ~ 9 standard deviations below random at $N=11$ (§Results, z_{plane}), and it is the same for every curve.

1.2.3. Exact expansion of the subspace counts

Let L have m elements, $p=m/u$, $e(x)=1_{L(x)}-p$ on $x\neq 0$, $e(0)=0$ ($\sum e=0$). For a k-dimensional V with $n_k=2^k-1$ nonzero points, $\prod_{x\in V^*}(p+e(x)) = \sum_{T\subseteq V^*}p^{n_k-|T|}e(T)$, with $e(T)=\prod_{x\in T}e(x)$. Summing over V and grouping the sets T of distinct nonzero elements by size t and rank r (the number of k-spaces containing T is $N_{k(r)} = \binom{N-r}{k-r}_2$):

$$A_k = \sum_t p^{n_k-t}\sum_{T,\,|T|=t} e(T)N_{k(\text{rank }T)}.$$

For a uniform random m-subset the same identity holds with $e(T)$ replaced by its expectation ε_t (it depends on t only), giving the exact E_k. Subtract, and use that $\sum_{|T|=t}(e(T)-\varepsilon_t)=0$ for every t (the elementary symmetric polynomial of the multiset of e-values is the same for the observed set and for a random one). Sets of t distinct nonzero vectors are independent for $t\le 2$, so with $D_c=\sum_{T\in c}(e(T)-\varepsilon_t)$ over the dependent classes c (rank $r_c<t$):

Proposition 2 (exact).

$$A_k - E_k = \sum_{t\ge 3}p^{n_k-t}\sum_{c\text{ dependent, size }t} D_c\big(N_{k(r_c)}-N_{k(t)}\big).$$

The smallest dependent classes are the **lines** $T=\{x,y,x+y\}$ ($t=3$, $r=2$) and the **planes** $T=\{x,y,z,x+y+z\}$ ($t=4$, $r=3$). Their centred correlations are (Fourier inversion, $\hat e$ the Walsh transform of e)

$$D_{\text{line}} = \frac{1}{6q}\sum_w \hat e(w)^3 - \frac{u(u-1)}{6}\varepsilon_3, \quad D_{\text{plane}} = \frac{1}{24}\left[\frac{1}{q}\sum_w \hat e(w)^4 - 3P_2^2 + 2P_4\right] - \frac{u(u-1)(u-3)}{24}\varepsilon_4,$$

with $P_j=\sum e^j$ (the bracket removes the degenerate solutions as above). Since $\hat e(w) = (W(w)+\sigma)/2$ for $w\neq 0$, $\hat e(0)=0$, $\sigma = s/u$, $s=W(0)=2m-u$, these reduce to the power sums of Proposition 1 plus terms of relative size $O(s^2/q)$:

$$D_{\text{line}} = \frac{C_3+3s-s^3/q}{48}+O(\sigma), \quad D_{\text{plane}} = \frac{R_4-3u^2}{384}+O(q) = -\frac{q^2}{384}(1+O(1/q)) \text{ for a curve.}$$

(i) k = 2: exact, one term. There are no classes with $t\ge 4$ when $n_2=3$, and $N_2(2)-N_2(3)=1-0$, so

$$A_2 = E_2 + D_{\text{line}}, \quad 48A_2 = u(u-1) + 3s(u-1) + 3(s^2-u) + C_3.$$

The term that carries the deviation from random is the 3-point correlation $C_3 = (1/q)\sum W^3$ alone. For a curve $C_3 = -\varepsilon(q-2)$, so $A_2/E_2 - 1 \approx -\varepsilon/q$: a part in 10^{-3} at $N = 11$, sign set by $\mathrm{Tr}(a)$. This is the alternating 1 ± 0.0005 seen in the parent data. It is not the 8 % effect.

(ii) k = 3. Two classes with the next largest coefficients,

$$A_3 - E_3 \approx p^4 D_{\text{line}}(N_3(2) - 1) + p^3 D_{\text{plane}}$$

(the omitted classes have $t \geq 5$). For $p = 1/2$ and a curve this gives $A_3/E_3 - 1 \approx -(7 + 7\varepsilon)/q$ (odd N).

(iii) k = 4. The same two classes, with $N_4(3) \approx q/8$ and $N_4(2) \approx q^2/96$:

$$A_4 - E_4 \approx p^{12} D_{\text{line}}(N_4(2) - N_4(3)) + p^{11} D_{\text{plane}}(N_4(3) - 1).$$

With $p = 1/2$, $E_4 = \binom{N}{4}_2 2^{-15}$, $\binom{N}{4}_2 \approx q^4/20160$:

$$\frac{A_4}{E_4} - 1 \approx \underbrace{-\frac{105}{q}}_{\text{plane: 4th moment}} \quad \underbrace{-\frac{35\varepsilon}{q}}_{\text{line: 3rd moment}} \qquad (N \text{ odd}),$$

and $-(105 - 35\varepsilon)/q$ for N even, where C_3 changes sign. The 4-point (plane) term is the mean shift and is the same for every curve; the 3-point (line) term moves the curve by $\pm 35/q$ with $\mathrm{Tr}(a)$ and, for random sets, is the largest source of scatter: z_{line} is $N(0,1)$ for a null, but for a curve it is the constant ± 0.41.

1.3. Results

All numbers are copied from `summary.json` / `rows.jsonl`. Populations: `orig` (the parent's 26 + 26 sets at $N = 9, 11$, 6 + 6 at $N = 13$), `rep` (fresh seed: 24 + 24, 4 + 4), `extra` nulls (150, 150, 40) and planted sets. $N = 12$ is a descriptive contrast.

1.3.1. P1: the 4th moment (exact)

Every curve at $N = 9, 11, 12, 13$ has exactly $\sum W^4 = 2q^3 - 2q^2 - 3q$ (one distinct value per N: 267909632, 17171474432, 137405386752, 1099377385472) and the matching $\sum W^3$.

N	curves, kurt	paired nulls, kurt (sd)	difference, 95 % CI	Cohen d	perm. p
9	2.003902405398264	3.017898522141077 (0.259)	1.01400 [0.9449, 1.0883]	5.54	1.0e-5
11	2.0009763236153613	2.979949697019886 (0.101)	0.97897 [0.9506, 1.0065]	13.65	1.0e-5
13	2.0002441257165615	2.9997707820642376 (0.051)	0.99953 [0.9700, 1.0300]	27.69	1.0e-5
12	2.0004882215871227	3.00965630381015 (0.064)	1.00917 [0.9705, 1.0443]	22.42	1.0e-5

The curve value has sd $\sim 10^{-16}$: all curves coincide. The Cliff's δ is 1 at every N.

1.3.2. Standardised correlations: how far below random

With $z_{\text{plane}} = D_{\text{plane}} / \left(\sqrt{N_{\text{planes}}}(p(1-p))^2\right)$ and the analogous z_{line}:

N	z_{plane} curves	z_{plane} nulls (sd)	z_{line} curves (range)	z_{line} nulls (sd)
9	−4.604	0.101 (1.014)	−0.442 to 0.445	1.009
11	−9.230	−0.084 (0.987)	−0.427 to 0.426	1.061
13	−18.472	−0.106 (0.986)	−0.407 to 0.417	1.093
12	−13.059	0.343 (1.280)	0.398 to 0.422	1.102

The curves' plane correlation is $\sim 0.204\sqrt{q}$ standard deviations below random (4.6, 9.2, 18.5) and varies by < 0.013 across curves. The line correlation is two-valued, ± 0.41, with sign $-\varepsilon$ (odd N).

1.3.3. The ratios and the first-order formula

N	curves ratio$_4$	paired nulls ratio$_4$	observed deficit × q	first-order × q	residual × q	pred/obs
9	0.7013	1.0190	152.9	135.4	17.5	0.885
11	0.9223	1.0164	159.1	112.2	46.9	0.705
13	0.9812	0.9942	153.9	105.1	48.9	0.683

12	0.9885	0.9983	47.2	67.6	−20.4	1.432

- The observed curve deficit is $\sim 155/q$ at all three prime N: it is a $1/q$ effect.
- The first-order prediction approaches its asymptote $105/q$ (the plane term; the line term averages to zero over $\mathrm{Tr}(a)$): 135, 112, 105.
- The residual is $\sim 48/q$ at $N = 11, 13$ (not $O(1/q^2)$). It is the same for both $\mathrm{Tr}(a)$ classes. Splitting by $\mathrm{Tr}(a)$ at $N = 11$: observed deficit 0.0939 ($a = 0$) and 0.0614 ($a = 1$), predicted 0.0722 and 0.0373. The split 0.0325 matches the predicted 0.0349.
- $k = 2$: $A_2/E_2 - 1 = D_{\mathrm{line}}/E_2$ exactly. Curves: 1 ± 0.0005 at $N = 11$ (sd 0.0005), $+1/q = +0.000242$ at $N = 12$ (all $\varepsilon = +1$ there).
- $k = 3$: curve deficit observed / first-order: 0.015141 / 0.015081 ($N = 9$), 0.003566 / 0.003517 ($N = 11$), 0.000853 / 0.000844 ($N = 13$); all-set R^2 0.965, 0.993, 0.998.
- Over nulls the $k = 4$ fit of observed on predicted has slope 0.947, $R^2 = 0.924$ ($N = 11$) and slope 0.986, $R^2 = 0.983$ ($N = 13$). Over curves alone: $R^2 = 0.903$ and 0.968. Figure 3 shows nulls on the diagonal and odd-N curves $\sim 48/q$ below it.
- The exploratory A_3-aware version (rank-3 block replaced by the measured A_3 deviation) raises the null R^2 to 0.986 ($N = 11$) and 0.999 ($N = 13$) but the curve deficit only from 0.0548 to 0.0555: the rank-3 classes are not what is missing. What is missing lies in rank-4 dependent classes ($t \geq 5$), which were not computed.

1.3.4. P2: correlation with the 4th moment, and what carries the scatter

N	pooled Spearman (95 % CI), perm. p	pooled Pearson	within nulls Spearman, p	within curves
9	0.444 (0.332, 0.544), 1.0e-5	0.480	0.195, 0.0029	undefined (sd 0)
11	0.435 (0.322, 0.536), 1.0e-5	0.589	0.029, 0.342	undefined
13	0.302 (0.036, 0.527), 0.0098	0.553	−0.118, 0.794	undefined
12	0.048 (−0.223, 0.313), 0.355	0.146	−0.061, 0.664	undefined

The pooled correlation is the arm difference. Inside the nulls the 4th moment does not move ratio$_4$ (except weakly at $N = 9$): within-null R^2 of ratio$_4$ on kurt is 0.0016 ($N = 11$), 0.0054 ($N = 13$). Instead the 3-point correlation does:

N	R^2 on z_{line} (nulls)	R^2 on z_{plane} (nulls)	sd ratio$_4$: curves / paired nulls
9	0.493	0.045	0.1456 / 0.2853 (0.51)
11	0.915	0.0016	0.0182 / 0.0474 (0.38)
13	0.976	0.0056	0.0048 / 0.0120 (0.40)

The curve/null scatter ratio (0.38, 0.40) equals the ratio of the z_{line} standard deviations (0.413 / 1.061 = 0.39; 0.426 / 1.093 = 0.39). **Reading:** the rigidity that reduces scatter is that of the Kloosterman 3rd moment, $C_3 = -\varepsilon(q-2)$, not of the 4th.

1.3.5. Positive control

A rank-one Walsh bias: $L = \{x \neq 0 : \ell(x) = 0\}$, each membership flipped with probability η (density stays 1/2). At $N = 11$, $\eta = 0.46, 0.42, 0.38, 0.34, 0.30$:

η	kurt (mean)	ratio$_4$ (mean)	first-order prediction (mean)
0.46	3.008	0.981	0.978
0.42	4.725	1.260	1.238
0.38	8.467	1.856	1.698
0.34	23.12	3.906	3.041
0.30	51.76	9.648	5.969

Spearman of kurt with ratio$_4$ over the 20 planted sets: 0.9444, 0.9985, 0.9940 at $N = 9, 11, 13$ (permutation $p = 1.0 \cdot 10^{-5}$). Direction as predicted. The first-order formula underestimates strong planting, as expected (non-perturbative regime). Unions of cosets of a subspace (exploratory, not gated) do not give a monotone relation: their counts are non-perturbative and the raw 4th moment alone does not determine the count.

1.3.6. Composite $N = 12$

The 4th moment is identical (2.0005), but $C_3 = +(q+2)$ flips the sign of the line term, so the first-order deficit is $-(105 - 35)/q \approx 0.0165$ against 0.0115 observed (curves 0.9885 against paired nulls 0.9983, $p = 0.0715$, 10 vs 10; null sd 0.020). The parent's "no difference" (6 vs 6) is consistent with a 1–2 % effect that the sample cannot resolve. The all-lift $\mathbb{F}_4$-subspaces of $\mathbb{F}_4$-dimension 2 (the subfield structure) number 85–131 per curve against a null mean of 3.22 (max 9), but are at most 0.026 %

of A_4. Removing them changes the Koblitz ratio from 0.9875974815932574 to 0.9873476161452932. **Subfield cancellation is not supported; the first-order formula, with its sign flip, already accounts for the small $N = 12$ effect.**

1.4. Figures

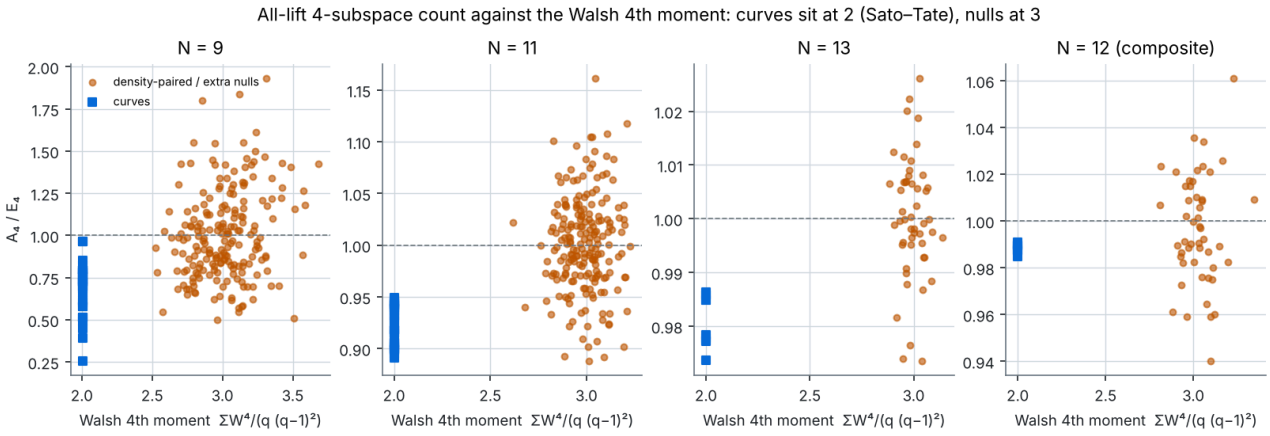


Figure 1: A_4/E_4 against the Walsh 4th moment. Curves (blue squares) share one 4th moment; nulls scatter around 3.

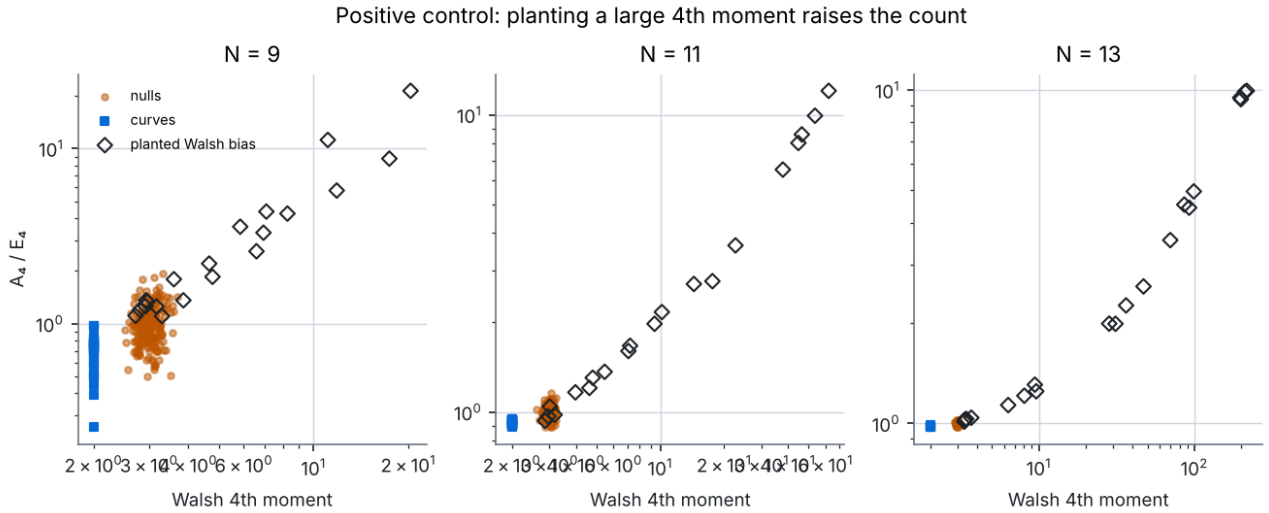


Figure 2: Positive control: planted rank-one Walsh bias raises both axes together (log–log).

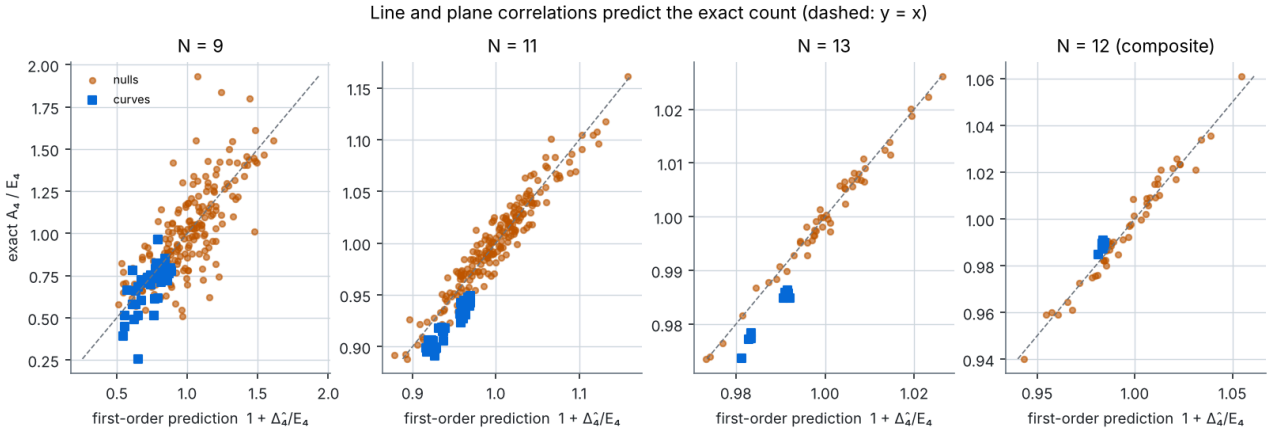


Figure 3: First-order prediction against the exact ratio. Nulls lie on the diagonal; curves at odd N lie $\sim 48/q$ below.

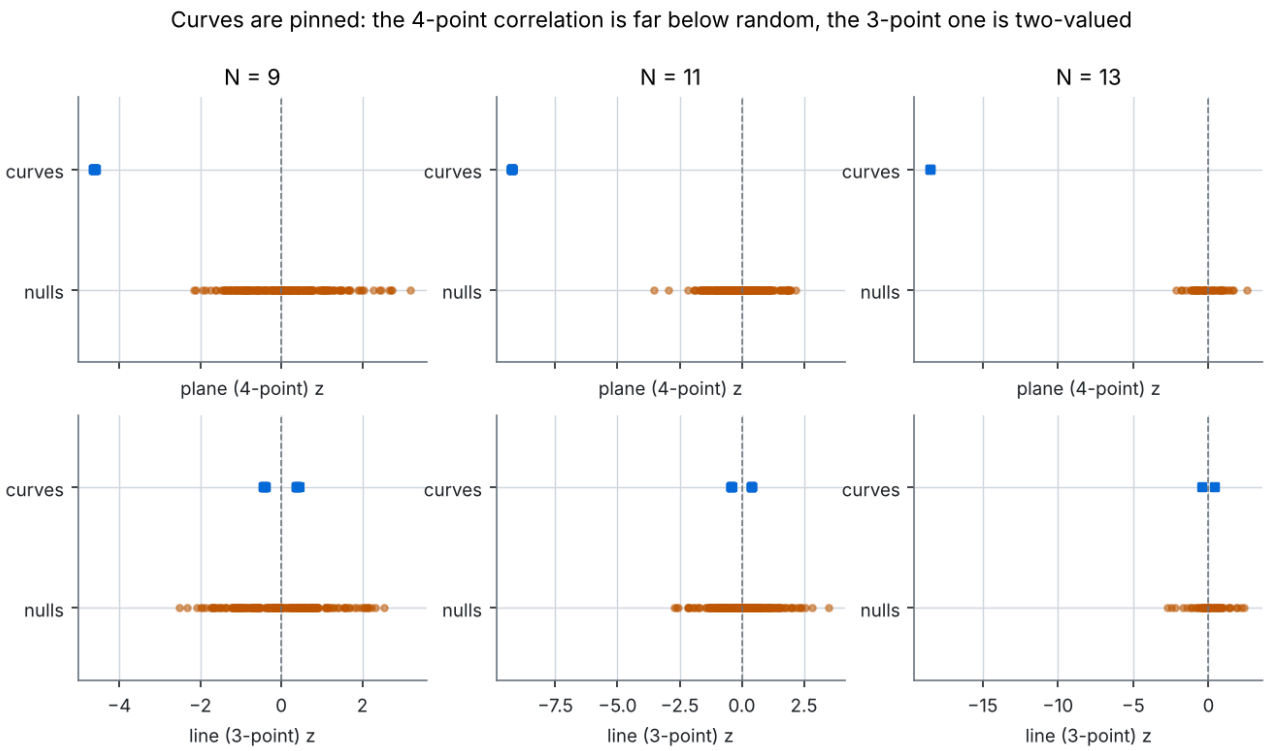


Figure 4: Standardised plane and line correlations. Curves: plane z far below random and constant, line z two-valued.

1.5. What is and is not established

- **Theorem** (proved above, checked on 120 curves): all Walsh power sums of a curve are curve-independent; $C_3 = -\varepsilon(q-2)$ (N odd), $C_{4'} = -(q-2)(q-4)$.
- **Exact computation:** $A_2 = E_2 + D_{\text{line}}$, the exact expansion of $A_k - E_k$ (Prop. 2).
- **Measured:** the first-order $k = 4$ formula explains 68–89 % of the odd-N curve deficit and the null scatter (R^2 0.92–0.98); line term confirmed by the $\mathrm{Tr}(a)$ split; scatter reduction carried by z_{line}.
- **Not established:** the remaining $\sim 48/q$ (rank-4 dependent classes, $t \geq 5$; the 5- and 6-point correlations are also curve-independent but were not evaluated); behaviour for $N > 13$, $k \geq 5$; any ECDLP cost.
- **Failed:** P2 within nulls (1 of 3 sets of N significant). The 4th moment alone does not drive the count of random sets.
- No ECDLP was solved or priced; no leaderboard, benchmark or rho row.

1.6. Reproduce

`cargo build --release && ./target/release/lift-repulsion-mechanism run reports/run-<UTC>/ --threads 3` (about 18 minutes, 748 sets), `python3 plot.py reports/run-<UTC>/`, then `python3 -c "import typst; typst.compile('REPORT.typ', output='REPORT.pdf')"`.